

Mishaal Hai

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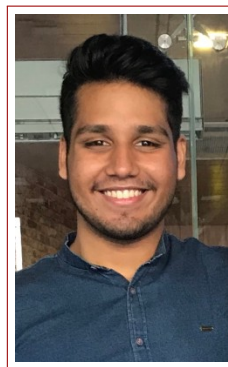
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Professional Experience

- November 2025 to Present **Research Assistant**, *Computer Science and Engineering Department*, Brac University, Dhaka, Bangladesh.
- April 2025 to Present **Program Coordinator**, *ICTP Physics Without Frontiers Bangladesh*.
- June 2020 to January 2023 **Adjunct Lecturer**, *Maths and Natural Sciences Department*, Brac University, Dhaka, Bangladesh.
- February 2020 to April 2020 **Physics Student Tutor**, *Maths and Natural Sciences Department*, Brac University, Dhaka, Bangladesh.

Education

- 2023–2025 **Master of Science in Theoretical Physics**, Università di Bologna, Bologna, Italy.
Thesis: Inflation models in type IIB string compactifications
- 2016–2020 **Bachelor of Science in Physics**, Brac University, Dhaka, Bangladesh.
Thesis: Non-Local Gravitational Interactions and the Black hole Information Paradox

Publication & ArXiv Pre-Prints

- D. Chakraborty, M. Hai, S. T. Jahan, A. R. Kamal, and M. S. J. Shuvo, “Effect of moduli redefinitions on fibre inflation”, *Journal of Cosmology and Astroparticle Physics*, **JCAP06 (2026)**, [arXiv: 2511.19610v3](#) [hep-th].
- M. Hai, A. R. Kamal, N. F. Shamma, M. S. J. Shuvo, “Perturbative Kähler Moduli Inflation”, [arXiv: 2506.08083](#) [hep-th]. **ArXiv Pre-Print** (Under journal review).

Relevant Graduate Courses Undertaken

- Advanced Mathematical Methods (Differential forms, Bianchi Universes, etc)
- Advanced Quantum Field Theory (Fadeev-Popov, ghosts, BRST Quantization methods, Background field method, Anomalies and QFT in Curved Spacetime).
- Supersymmetry.
- Standard Model.

Research Interests (in no particular order)

- Quantum Field Theory
- The Standard Model and BSM.
- String Theory and String Phenomenology.
- Black hole Thermodynamics.
- Cosmology

Technical Skills

- Proficient in Mathematica.
- Intermediate level coding in Python.
- Intermediate level video editing in Adobe Premiere Pro.

Extracurricular/Academic Activities

- Grant recipient of the RSGI 2025. Grant ID: BRACURSGI25014; **Time-dependent Backgrounds in String Theory and Early Universe Cosmology**.
- Co-Principal Investigator for the RSGI 2025 grant-supported project.
- Program Coordinator of ICTP's Physics without Frontiers, Bangladesh. We organise, under Prof. Nabil Iqbal's supervision, school's, workshop's, research internship's, etc.
- I had been the teaching assistant in the Quantum Field Theory school, as well as the General Relativity school in ICTP's Physics Without Frontiers Bangladesh.
- Attended ICTP's *String Math 2024* conference in Trieste, Italy.
- Attended the *Deconstructing the String Landscape (a.k.a. landscapia) 2023* conference in Paris, France.
- I am one of the founding members and organizers of *SPARQ* (Strings, Particles, Astrophysics, Relativity and Quantum Computing.) This was an interactive *technical talk* based workshop, where students and professionals are given a platform to present their current work.
- Involved in independent filmmaking, e.g. [A Beautiful Curse](#).

Talks Given

- Presented Stephen Hawking's work in "Particle Creation by Black holes" and the viability of Non-thermal radiation as a resolution of the information paradox in the SPARQ workshop.
- Presented a review of Suvrat Raju and Kyriakos Papadodimas's "Black hole Interior

in AdS/CFT and the Information Paradox" in the SPARQ workshop.

- I presented our paper on "Perturbative Kähler Moduli Inflation" in Merul Badda Talks, Dhaka, Bangladesh.